

iSHAP: Rule-Guided Synthetic Data Augmentation for Better Feasibility–Performance Trade-offs

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Abstract

We present a practical workflow for steering model behavior with lightweight structural rules while preserving predictive performance. The key idea is to iteratively generate synthetic training points and optimize them against a combined objective: hold-out task performance (root mean squared error for regression, accuracy for multi-class classification) and a feasibility score grounded in *do*-interventions on the fitted model, not correlation with SHAP values alone. Unlike post-hoc explanation-only methods, our approach actively edits training data to align predicted interventional effects with user-specified directional rules. We provide an implementation based on LightGBM, Optuna (TPE), and TreeSHAP only for optional plots, and demonstrate the pipeline on two tabular datasets.

1 Introduction

Many machine learning systems achieve strong predictive metrics but violate simple domain expectations. For example, in glucose forecasting one may expect that increased insulin should decrease future glucose, yet unconstrained models may learn unstable or non-monotone interventional curves. This creates a gap between pure predictive optimization and rule-consistent behavior.

We propose iSHAP, a method that uses structural rules and model-based *do*-effects to guide synthetic augmentation. At each round, candidate synthetic points are optimized with respect to a combined objective balancing:

1. prediction quality on hold-out data (task-appropriate metric), and
2. rule feasibility from interventional summaries of the model response.

If a candidate improves the combined objective, it is added to training data and the model is retrained. This creates a closed loop between model fit, interventional diagnostics, and data augmentation.

2 Method

2.1 Problem Setup

Given training data (X, y) and directed rules $\mathcal{R} = \{(u \rightarrow v, s)\}$ with desired qualitative sign $s \in \{-1, 0, +1\}$ (negative, none, positive), we train a predictive model and evaluate rule consistency under interventions implied by the model map.

Rules to the target. For a rule $(u \rightarrow y)$, fix a validation covariate matrix X and sweep a grid g over the observed range of feature u . Define the interventional prediction curve

$$\bar{f}(g) = \frac{1}{n} \sum_{i=1}^n f(X_{-u}^{(i)}, u=g),$$

where row i is copied and only column u is set to g . We summarize alignment with the desired monotone direction (higher u should raise or lower $\bar{f}$ depending on s) using a monotone score derived from the consecutive differences of $\bar{f}(g)$ along the grid (full agreement with the desired direction yields the strongest feasibility contribution). A Pearson correlation between g and $\bar{f}(g)$ is retained as a diagnostic correlate of global slope but feasibility optimization emphasizes monotone agreement with the rule sign.

Rules between features. For $(u \rightarrow v)$ with both u, v among inputs, we use the partial correlation between X_u and X_v after linearly controlling for all other observed features—a compact backdoor-style adjustment when the full causal graph is unknown.

Aggregating weighted deviations of each rule’s summary from the ideal sign yields a feasibility score in $[0, 1]$.

2.2 Optimization Objective

At each round, Optuna (TPE sampler) proposes synthetic points. Each trial is scored by a combined objective that rewards better hold-out performance and higher feasibility. The trade-off is controlled by `objective_metric_weight` in code:

- 0.0: prioritize hold-out performance only (regression: lower root mean squared error; classification: higher accuracy),
- 1.0: prioritize feasibility only,
- intermediate values: linear trade-off in the implemented scalar.

2.3 Round-Based Augmentation

The pipeline:

1. Train baseline model on current augmented set.
2. Optimize a batch of synthetic points.
3. Evaluate combined score on fixed validation diagnostics.
4. Accept batch if combined score improves; otherwise stop.

3 Experimental Setup

3.1 Datasets

We currently evaluate:

- **d1namo**: glucose-related tabular *regression*; hold-out metric is root mean squared prediction error (reported as `error` in outputs).

- **cancer_air_pollution**: cancer risk *classification* with ordinal labels mapped to classes; hold-out metric is classification *accuracy*.

3.2 Rules

For controlled sanity checks, we use one rule per dataset:

- `dlnamo`: `insulin` $\rightarrow$ `future glucose`, negative (higher insulin should correspond to lower predicted future glucose along the do-curve).
- `cancer_air_pollution`: `Air Pollution` $\rightarrow$ `Level`, positive.

4 Results

4.1 Quantitative Summary

Table 1 summarizes a recent pipeline run exported to `results/<dataset>/feasibility.json`. Performance and feasibility definitions differ by task as described above; the feasibility column aggregates interventional alignment for target rules (and partial correlations for feature-feature rules when present).

Table 1: Before/after summary (representative run; re-run scripts to refresh numbers).

Dataset	Hold-out metric	Before	After	Δ Feas.	Synth rows
<code>dlnamo</code>	RMSE (error)	28.328	28.575	+0.198	80
<code>cancer_air_pollution</code>	Accuracy	1.000	1.000	+0.137	60

For `dlnamo`, feasibility optimizes a *monotonic signed* summary for target rules (`feasibility.json` `pearson_r_*` rows store that score; key name is historical). Pearson ρ between grid and curve (`do_effect_corr_*` in the feasibility blob; `before_do_corr` / `after_do_corr` in the do-curve file) may fall while feasibility still climbs. Archived JSON carries both `monotonic_signed_score`-style summaries and step-alignment fractions. For `cancer_air_pollution`, the rule summary moved from 0.720 toward 0.993 while hold-out accuracy remained at perfect accuracy on the evaluated split.

4.2 Visual Evidence: SHAP and interaction views

Figures 1–4 show before/after SHAP beeswarm and correlation-based interaction networks for context. They are not the primary feasibility signal; the interventional curves below are.

4.3 Interventional (*do*) curves

Figure 5 and Figure 6 plot $\bar{f}(g)$ versus intervention level g for each target-directed rule before and after augmentation. Numeric grids and predictions for the plotted curves live only under `results/<dataset>/` as `do_curve_before_after.json` (long CSV alongside). Metrics there use the same intervention grid subsample as `compute_feasibility` (avoid comparing a finer grid to coarse optimization summaries).

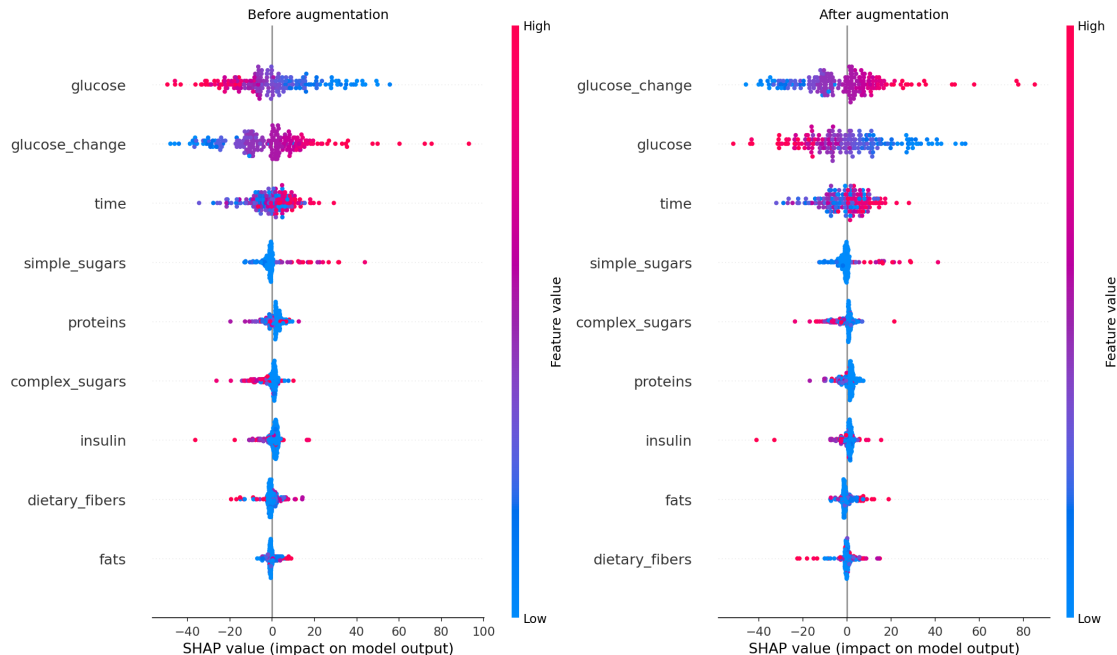


Figure 1: dlnamo SHAP beeswarm before and after augmentation.

5 Discussion

Interventional feasibility makes the notion of “insulin lowers future glucose” directly testable on the fitted predictor, independently of beeswarm correlations. Splitting hold-out metrics by task (RMSE vs. accuracy) avoids mis-stating performance on the cancer benchmark. The stored do-curve tables allow external reproduction and alternative plot styles without re-fitting.

Current limitations include dependence on the choice of validation rows for the do-sweep, grid resolution, and the strength of partial-correlation surrogates for non-target rules when the true graph is unknown.

6 Limitations and Future Work

Future work should investigate richer SCM-based adjustment sets for feature–feature rules, adaptive grids, and joint multi-rule interventional constraints.

7 Conclusion

iSHAP couples data-centric synthetic search with interventional and partial-correlation feasibility so practitioners can trade predictive metrics against rule-shaped model behavior with auditable curves and saved numeric artifacts.

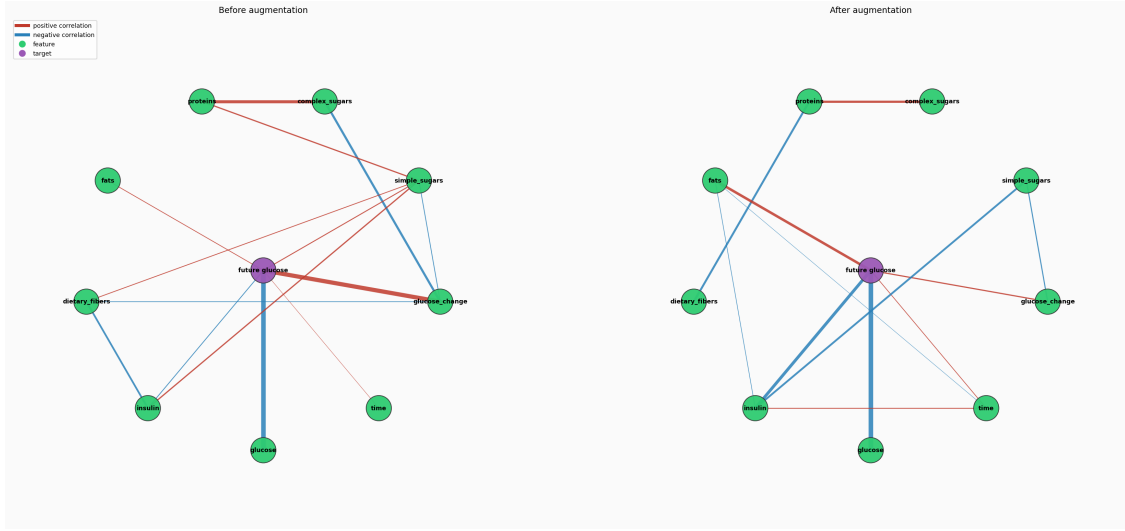


Figure 2: d1namo correlation-based interaction network before and after augmentation.

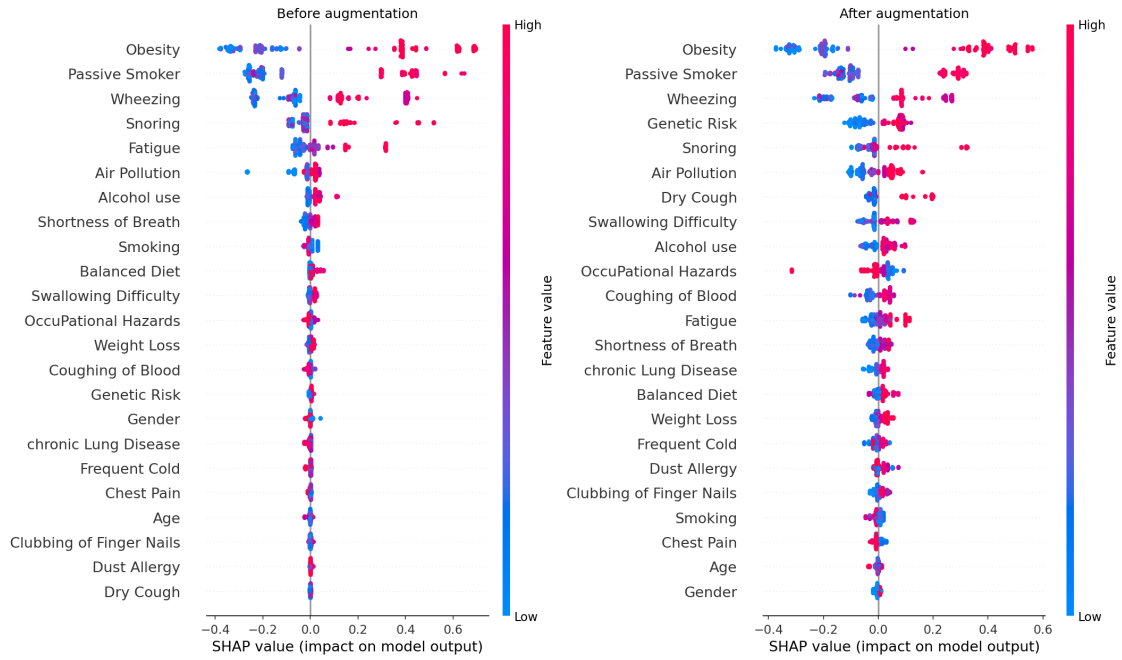


Figure 3: cancer_air_pollution SHAP beeswarm before and after augmentation.

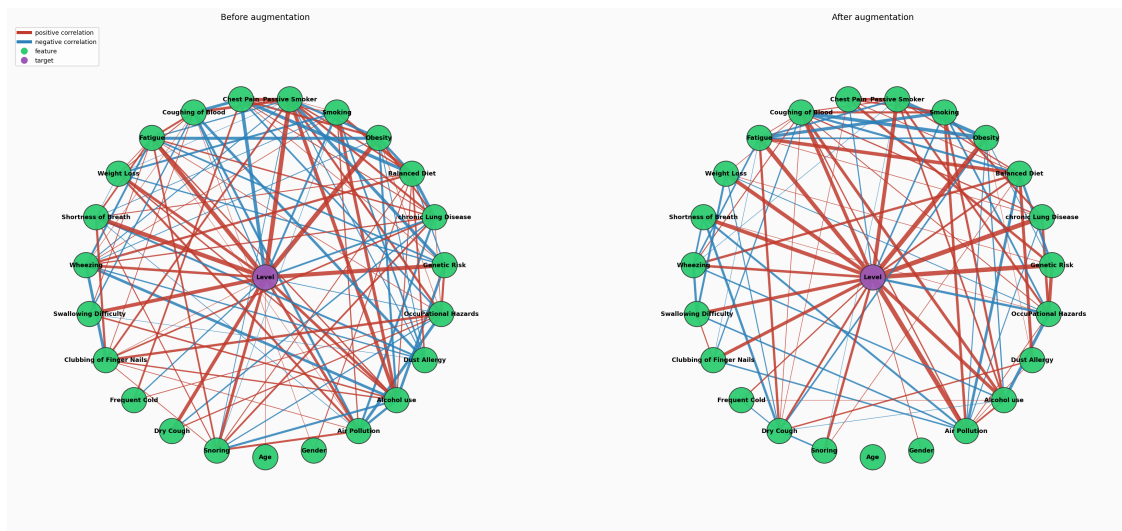


Figure 4: cancer_air_pollution correlation-based interaction network before and after augmentation.

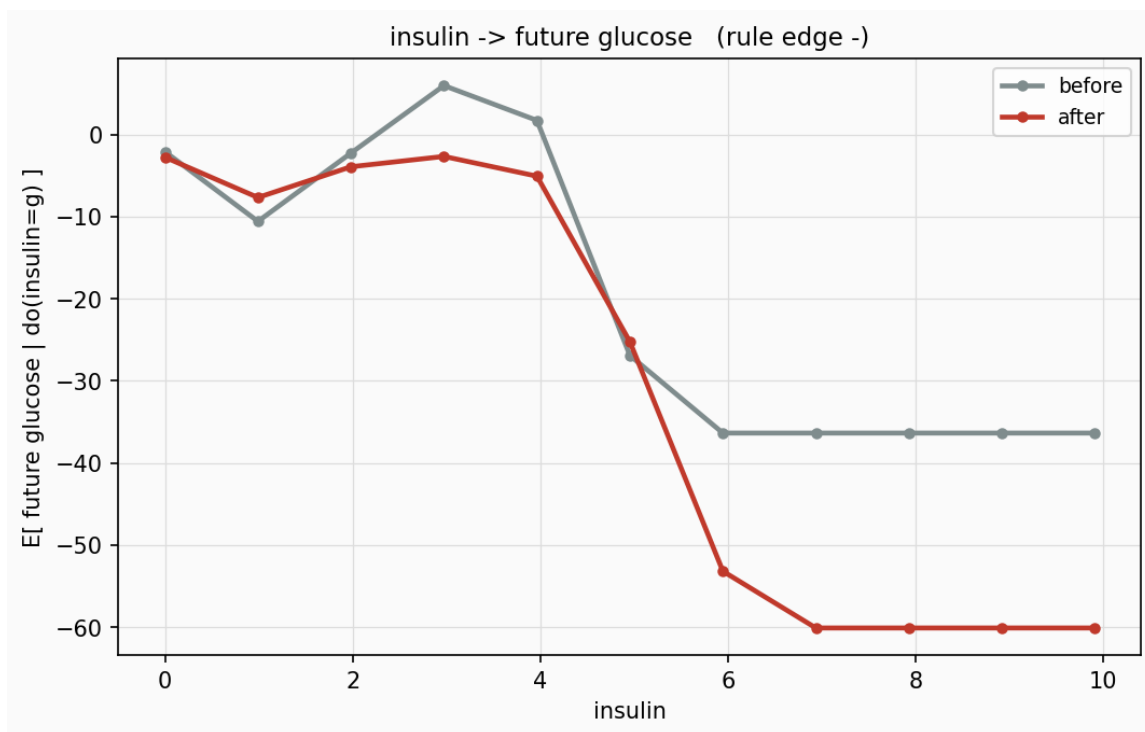


Figure 5: dlnamo: interventional mean prediction vs. fixed u for the insulin rule (before vs. after augmentation).

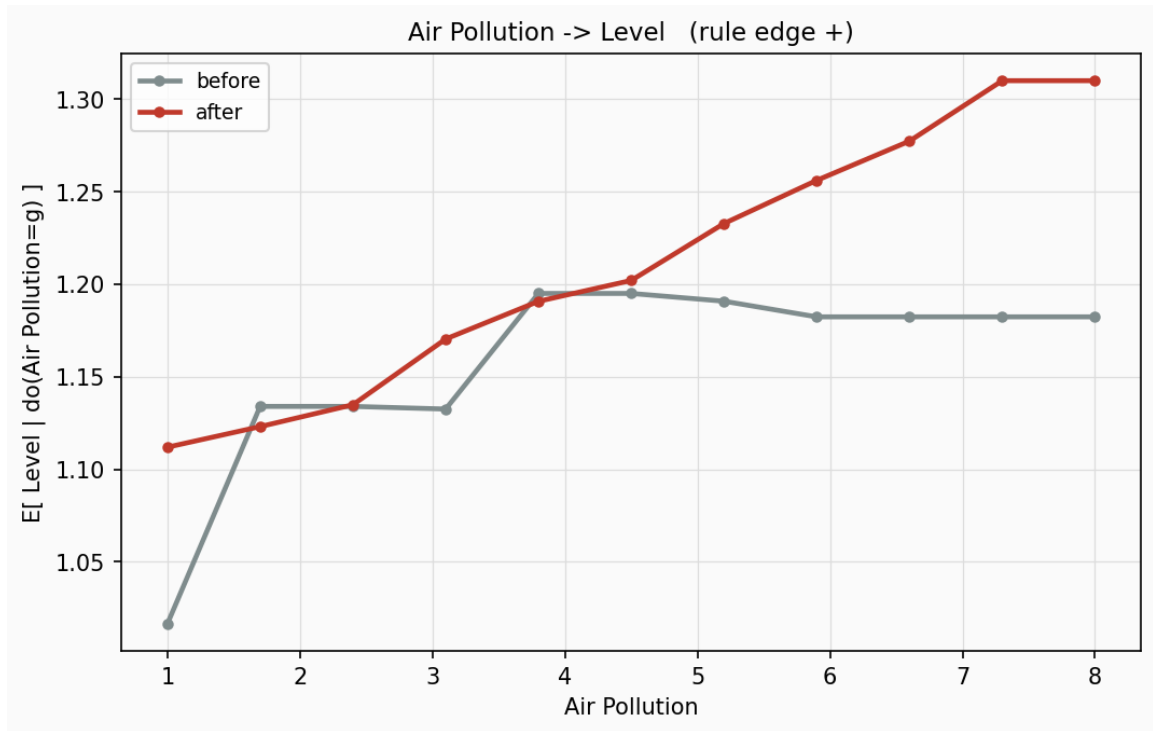


Figure 6: cancer_air_pollution: same construction for Air Pollution $\rightarrow$ Level (classification model; $\bar{f}$ is mean predicted score / class logits aggregated as in the codebase).